

Convertibility Buys Causality, Topology Buys Meaning: Weight-Based Mechanistic Interpretability for Bilinear Networks

What tensor-network decomposability does, and does not, give you

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Abstract

A growing line of work replaces every nonlinearity in a network with bilinear / polynomial operations, so that the entire learned map becomes a single tensor network that can be decomposed directly from its weights. The implicit promise is that this decomposability yields interpretability. We show that promise splits into two properties that are usually conflated, and that only one of them is actually free. **(1) Convertibility buys causality.** Exact tensor-network export gives a weight-derived mechanism basis whose global orthogonalize–diagonalize–truncate (ODT) ranking is *causally faithful*: on image classifiers and a softmax-free vision transformer, keeping the top-ranked directions preserves accuracy and removing them destroys it — *matching* an unsupervised PCA baseline and far exceeding random on deletion/insertion faithfulness, while being weight-derived and globally rankable; this structure survives through attention (via a data-driven Gram at attention-crossing bonds) and, in our small models, increasing depth. **(2) Topology buys meaning, and it is not free.** Whether those causal directions are *human-coherent* (localized, nameable) is governed not by convertibility but by the network’s *topology*: architectures that preserve spatial locality yield atoms that reach $\sim 3.5\times$ random-baseline locality — $\sim 1.5\text{--}2.3\times$ more localized than dense — at no accuracy cost, robust across three seeds and two datasets, and (in a deeper replication) an effect that *strengthens* with depth. **(3) For a policy, mechanisms are built by attention and readable only at the right bond.** On a tensor-decomposable vision-language-action policy (χ -VLA, which we build as the vehicle), the action-relevant quantity (which object the instruction names, and where it is) is not linearly present at the network input ($R^2 \approx 0.2$) but becomes decodable after the attention that constructs it ($R^2 \approx 0.9$); moreover linear decoding there is *not* causal, reinforcing why weight-derived causal bases matter over probing. Together: convertibility guarantees a causal, rankable basis; *meaning* must be engineered through topology and bond placement. We give the design implications for scaling interpretable models.

1 Introduction

Interpreting a large policy is hard largely because the network is not compositional. Nonlinear operations such as the softmax in attention, the smooth activation in an MLP, and layer normalization prevent us from rewriting the model as a single algebraic formula, so we are forced to study its behaviour on data rather than its weights directly.

A recent line of work argues that this is a design choice, not a necessity. If every learned operation is kept polynomial in the activations, meaning it is built only from linear maps, element-wise multiplications, additions, fixed masks and permutations, learned constants, and fixed scalar or diagonal rescalings, then the model is a structured polynomial. Once token identities are written as one-hot vectors, that polynomial is a single multilinear object, commonly called a tensor network, and tensor networks can be decomposed, ranked, and truncated using tools from numerical linear algebra applied to the weights alone.

This paper asks the question the line of work leaves implicit: *does decomposability actually buy interpretability?* We find it buys exactly half. Our contributions are three, established across three substrates of increasing difficulty (image classifiers, a softmax-free vision transformer, and a full tensor-decomposable policy):

1. **Convertibility buys causality, for free.** The exact weight-derived ODT basis has a global importance ranking that is causally faithful by deletion/insertion — *matching* unsupervised PCA and far exceeding random, while being weight-derived, exact, and globally rankable — and the structure holds through attention (data-drivenly) and, in our small models, suggests growing compressibility with depth.
2. **Coherence is bought by topology, at no capability cost.** Whether the causal directions look like recognizable features is set by architecture, not convertibility: spatial-locality-preserving topology reaches $\sim 3.5\times$ random-baseline locality ($\sim 1.5\text{--}2.3\times$ more localized than dense), robust across seeds, datasets, and depth, and the most coherent model is also the most accurate.
3. **For a policy, mechanisms live behind the attention.** The action-relevant quantity is constructed by attention and readable only at a post-attention bond; and decoding it there is not the same as controlling it.

Relation to prior tensor-interpretability work. Dooms et al. [3] give tree-structured ODT as a *compression* result; Pearce et al. [2] and Déhérand [4] *assert* that architecture affects the coherence of bilinear eigen-features on shallow convolutional models, without an intervention. Our delta is to (i) separate *causal* from *coherent* as an explicit, tested axis with interventions and standard faithfulness baselines; (ii) establish topology as the coherence lever at matched capability, with depth-scaling and cross-dataset de-risking; and (iii) introduce the *bond-placement* principle for policies. The vehicle for the policy substrate is χ -VLA, a Vision-Language-Action model (mapping camera images and a language instruction to robot actions) that we build to be fully tensor-decomposable end to end, along the modular OpenVLA [1] skeleton, whose four learned stages each have a published polynomial replacement:

- **Bilinear feed-forward layer** (Riggs [2]). Replace the gated MLP $D(\text{swish}(Lx) \otimes Rx)$ with the activation-free $D[(L\bar{x}) \otimes (R\bar{x})]$, where $\otimes$ is the element-wise (Hadamard) product and $\bar{x}$ is defined below. This is exactly a low-rank factorization (a CP, or canonical polyadic, factorization) of a third-order weight tensor.
- **Softmax-free bilinear attention** (Riggs [2]). Replace the softmax attention pattern with the element-wise product of two separate learned query-key score matrices.
- **Foldable normalization** (Dooms et al. [3]). Normalize during training by dividing by a running average magnitude (a single scalar), then after training freeze that scalar and multiply it into the next weight matrix, so the deployed model contains no division at all.

- **Linear action head.** Replace the discretized action tokens and language-model-style decoding with a direct linear map from learned action-query states to a continuous action vector.

We assemble these along the OpenVLA skeleton, keeping only genuinely non-learned interfaces outside the tensor-network boundary (tokenization, camera decoding, safety clipping, motor communication), so that χ -VLA’s learned map from pixels and token identities to a continuous action is one tensor network. The architecture is the substrate, not the contribution: it lets us ask what tensor-decomposability buys for interpretability on a real policy, not just a classifier. We scope the policy claims to the *readability and faithfulness of action-relevant mechanisms*; closed-loop robot control is out of scope (we report offline behaviour and grounding only).

2 Background: the three polynomial primitives

The homogeneous coordinate. Throughout, we write $\bar{x} = [1; x]$ for the input with a constant 1 prepended. This “homogeneous coordinate” lets a bilinear layer represent not only quadratic terms but also linear terms and constants (a bias becomes the weight on the constant coordinate). Without it, a stack of L multiplicative layers would be forced to produce only degree- 2^L terms; with it, the network can represent every degree up to 2^L , including the near-linear functions that most of a real task needs.

Bilinear feed-forward layer (a CP-factorized cubic tensor). With $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \otimes (R\bar{x})], \quad y_o = \sum_{i,j} \left(\underbrace{\sum_r D_{or} L_{ri} R_{rj}}_{T_{oij}} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The middle expression makes the tensor structure explicit: the layer is exactly a third-order weight tensor $T \in \mathbb{R}^{d \times (d+1) \times (d+1)}$ contracted twice with the input. The factored form uses about $3dr$ parameters instead of the d^3 of the dense tensor, which is why it is called a CP factorization (a sum of r rank-one terms).

Softmax-free bilinear attention. Each head forms two query-key systems $A_1 = Q_1 K_1^\top$ and $A_2 = Q_2 K_2^\top$ and one value matrix V . The attention pattern is their element-wise product, with no softmax:

$$Y = C[M \otimes (A_1 \otimes A_2)/s]V. \quad (2)$$

Here M is a fixed mask (for example, lower-triangular for causal models), C is a fixed per-row rescaling $C_{ii} = 1/\sqrt{n_i}$ with n_i the number of visible keys, and s is a fixed scale. Because there is no softmax, the whole operation is polynomial in the activations. It can be evaluated three mathematically identical ways: the explicit $N \times N$ pattern; a “Khatri-Rao” form that avoids the $N \times N$ matrix by building row-wise Kronecker features; and a causal prefix-scan. We verified all three agree to 10^{-5} in single precision.

Foldable normalization. During training we divide activations by a single running scalar equal to their average root-mean-square magnitude. After training we re-estimate that scalar on a calibration set, freeze it, and multiply it into the following weight matrix ($W' = W/c$ before a linear map, or $L' = L/c$, $R' = R/c$ before a bilinear layer). A frozen scalar division is a fixed rescaling, so the deployed model has no operation whose divisor depends on the input. This is what we mean by “fold”: the normalization module disappears into the weights.

Ordinary component	χ -VLA replacement	Algebraic form
Smooth activation (ReLU, GELU)	element-wise product of two linear maps	$(Lx) \otimes (Rx)$
Gated MLP	bilinear (CP) feed-forward	$D[(L\bar{x}) \otimes (R\bar{x})]$
Softmax attention	product of two score matrices	$C[M \otimes (A_1 \otimes A_2)/s]V$
Layer/RMS normalization	foldable scalar normalization	x/c with c a frozen scalar
MLP projector	cross-bilinear (CP) projector	$W_s s + W_m m + D_p[(L_s \bar{s}) \otimes (R_m \bar{m})]$
DINOv2 / SigLIP encoders	two polynomial χ -ViTs	(all blocks bilinear)
Llama 2 backbone	polynomial transformer	(all blocks bilinear)
Discrete action tokens	linear continuous-action head	$\hat{a}_k = W_a h_k + b_a$
Bias terms	constant (homogeneous) coordinate	$\bar{x} = [1; x]$

Table 1: Component-level replacements. A bar denotes the homogeneous coordinate $\bar{x} = [1; x]$; $\otimes$ is the element-wise product.

3 From OpenVLA to χ -VLA

Figure 1 contrasts the two pipelines and Table 1 lists the block-level replacements. The high-level flow is deliberately preserved, while every learned internal block is replaced by a polynomial one.

Foldability is achievable but requires care (details in Appendix A). Making the scalar (foldable) normalization stable inside a deep degree-two stack requires keeping residual branches near the identity (small learned gains, initialized at 0.01) and distilling from a stable per-token teacher; the resulting student matches its teacher and folds to a normalization-free graph to 10^{-6} (fp32) / 10^{-12} (fp64). This is an engineering prerequisite for the substrate, orthogonal to the interpretability findings that follow.

4 Exact global decomposition (ODT), in depth

This section is the payoff of the whole design and we treat it carefully. An ordinary softmax-and-normalization policy cannot be turned into a single algebraic object, so it cannot be globally diagonalized; a tensor network can.

4.1 What ODT is, and why it needs a tensor network

Suppose we want to compress a network by discarding directions in one of its hidden spaces, keeping only the ones that matter for the output. Call the hidden space we are analyzing a “bond”. The naive approach is a local one: take the weight matrix that reads from that bond, run a singular value decomposition (SVD), and keep its top directions. This is greedy and per-layer; it only sees the one adjacent weight matrix, not what the rest of the network does with those directions downstream.

Orthogonalize-Diagonalize-Truncate (ODT), introduced for tree-shaped bilinear networks by Dooms et al. [3], instead ranks bond directions by their *global* importance, accounting for the entire rest of the network, which we call the “environment” of the bond. This is only possible if the whole model is one multilinear object, because only then can the environment be contracted into a single operator whose singular directions are the true global importances. The three steps are:

1. **Orthogonalize.** Sweep through the network and replace each weight core by an isometry (an orthonormal map) times a small triangular remainder, using a reduced RQ decomposition

of the matricized core, pushing the remainder toward the output. After this sweep each bond has an orthonormal basis, so importance can be read off directly as singular values rather than being tangled up with the basis.

2. **Diagonalize.** At a bond, form the Gram matrix of the environment, $G = Q^*Q$, where Q is the downstream map from the bond to the output. Its eigenvalues are the squared global importances of the bond directions and its eigenvectors are those directions.
3. **Truncate.** Keep the top k eigenvectors, insert the projector $P_k = V_k V_k^\top$ at the bond, and absorb it into the adjacent cores. The output error is bounded by the square root of the sum of the discarded eigenvalues, so the spectrum tells you in advance how much you can remove.

4.2 How we compute it for a bilinear chain

Our exact-decomposition model is a feed-forward bilinear network we call a χ -MLP, chosen because a chain with no attention and no residual connection is a clean tree, exactly the setting ODT was designed for. After folding, it is

$$z_0 = [1; Ex], \quad a_i = C_i(z_{i-1}, z_{i-1}), \quad z_i = [1; a_i], \quad \text{logits} = W_{\text{head}} a_L + b, \quad (3)$$

where E is a linear input embedding, each $C_i \in \mathbb{R}^{d \times (d+1) \times (d+1)}$ is a symmetric bilinear core recovered from the folded layer, and $C_i(z, z)$ means contracting the core twice with the homogeneous activation z .

To rank the directions of a bond a_i by global importance we build the exact downstream map from a_i to the output and contract it against itself over the output index and over every environment leg, leaving the bond leg open on both sides. For the deepest bond (the output of the first layer, whose downstream is two more bilinear layers) the downstream is a degree-four form in z_i . We construct it explicitly by composing the cores,

$$D[c, i, j, k, l] = \sum_{p, q} C_L^{\text{head}}[c, p, q] Z_2[p, i, j] Z_2[q, k, l], \quad G[i, i'] = \sum_{c, j, k, l} D[c, i, j, k, l] D[c, i', j, k, l], \quad (4)$$

where Z_2 expresses the second-layer homogeneous activation as a quadratic form in z_i and C_L^{head} folds the linear head into the last core. The $d \times d$ block of G over the non-constant coordinates gives the global importances. The local baseline uses only the adjacent core (the layer that directly reads the bond), ignoring everything downstream. Truncation projects the activation onto the top k eigenvectors and we measure the resulting accuracy.

4.3 Results on the χ -MLP

We train a three-layer χ -MLP (bond width 32) on the SVHN digit dataset to 0.81 accuracy and run the exact procedure.

- **Exact reconstruction.** The unrolled tensor network reproduces the trained module to 5×10^{-12} in double precision, confirming that the exported cores are the model, not an approximation of it.
- **Global low-rank structure.** Ranking the deepest bond by the global Gram, 20 of its 32 hidden dimensions (62%) can be removed with at most a 1% accuracy drop. Dooms et al. report about 70% on their own SVHN network; we report our measured figure rather than assuming their number transfers.

- **Global beats local.** At a matched number of kept directions, the global ranking preserves accuracy better than the local per-layer SVD, most clearly at the compression elbow.

retained directions k	4	6	8	12	16	
global ODT	0.571	0.777	0.797	0.809	0.811	(full width 32 gives 0.808; deepest bond)
local SVD	0.523	0.686	0.798	0.807	0.811	

The reason global wins is exactly the reason it needs a tensor network: at $k = 6$ the global ranking already knows which directions the two downstream layers will actually use, while the local ranking only sees the immediately following layer and keeps directions that later turn out not to matter.

4.4 From compression to interpretation: are the ranked modes real?

Removable-dimension counts show that low-rank structure *exists*; they do not show that the surviving directions are human-interpretable, nor that the ranking is *causal* rather than merely a good compressor. Déhérand [4] takes the complementary step, extending the χ -net idea to convolutional networks and reading each class output as a quadratic form whose eigenvectors, projected to pixels, are spatially coherent digit strokes. Two things are left open there: the interpretation is run only on a shallow single-bilinear layer, and visual coherence is never tied to an intervention. We close both on our own models.

Output-conditioned atoms. For a single-bilinear χ -classifier each class logit is *exactly* a quadratic form in the homogeneous embedded input, $\ell_c(x) = z^\top Q_c z + b_c$ with $z = [1; Ex]$ and $Q_c = \sum_o W_{c,o}^{\text{head}} C_o$ the head folded into the core (reconstruction against the trained module is 2.6×10^{-13} on SVHN, 0.80 accuracy). Eigendecomposing $Q_c = \sum_i \lambda_i v_i v_i^\top$ writes the logit as $\sum_i \lambda_i (v_i \cdot z)^2 + b_c$: signed “atoms” where $\lambda_i > 0$ supports class c and $\lambda_i < 0$ suppresses it, each projected to a pixel sensitivity map through E .

These atoms (Figure 2) are the object analyzed in Section 5, where we ask the three questions this section motivates: is the ranking *causal* (deletion/insertion faithfulness against integrated gradients and PCA), are the atoms *coherent* (a function of topology, not convertibility), and — for a policy — at which *bond* is the mechanism readable and controllable. We turn to those results next.

4.5 Does this survive attention and residual connections?

The χ -MLP is feed-forward; a transformer adds attention (which mixes information across token positions) and residual connections. Extending exact, weight-only global ODT to a general attention network is an open problem, because the attention pattern is itself a learned degree-four function of the inputs, so the environment is no longer a fixed operator. We give the first empirical evidence that the structure is nonetheless there.

We train a strict, foldable transformer language model (using the teacher-then-distill recipe so it is stable) and analyze the bond at the block input, whose downstream *crosses* the attention layer (each token feeds the queries, keys, and values of every position), then the residual path, the feed-forward layer, and the head. Because the attention pattern is data-dependent here, we rank directions by a data-driven version of the global Gram, the average outer product of the output gradient at the bond, which measures how much each direction of the bond actually affects the loss through the full downstream including attention. The local baseline is the SVD of the adjacent input projections (the query, key, value, and feed-forward input weights).

	retained k	global (output-aware)	local (weight SVD)
1 attention layer (full-rank loss 5.23)	32	6.11	7.88
	64	5.47	6.27
	96	5.27	5.55
2 attention layers (full-rank loss 5.98)	32	6.18	7.78
	64	6.02	6.52
	96	5.99	6.18

The global ranking beats the local one at every retained width for both models, and the attention model still folds to an exact tensor network (double-precision fold error at machine precision). Notably the two-layer model is *more* compressible (64 of 128 directions, 50%, removable at under 0.05 nats) than the one-layer model (25%), which suggests deeper attention stacks have more, not less, global low-rank structure. Two honest caveats. First, this uses the data-driven global Gram; the fully weight-based, data-free contraction across an attention-crossing bond, and the transfer of the tree error bound, remain open. For bonds whose downstream is feed-forward (for example, the bond after the attention sub-layer), the exact weight-based procedure of the previous subsection applies directly. Second, the strict scalar-normalization model needs the distillation recipe to train stably beyond one layer.

4.6 Cost and limits

For the tree network the procedure costs about $O(Lh^4)$ time with $O(h^3)$ memory in the bond width h , which is why our exact-decomposition models use a small width. Unrolling a full attention transformer into one tensor over all token positions grows combinatorially, so exact whole-model contraction is only feasible at small scale; the practical path is to decompose one analysis bond at a time, as above.

5 What decomposability buys: causality, then topology, then bonds

The previous section established that the exported bilinear network can be exactly decomposed and that a global ODT ranking is meaningful. Here we ask the sharper question: *how much of “interpretability” does that decomposability actually buy?* It buys a causal, rankable basis — but coherence and nameability are bought separately, by topology and by bond placement.

5.1 Causality is competitive with standard attribution, and (at feed-forward bonds) free

We compare the ODT direction ranking against real baselines — unsupervised PCA of activations, integrated gradients [12], and random — using deletion/insertion faithfulness AUC (project the input/bond onto, or out of, the top- k directions and measure accuracy). On the softmax-free χ -ViT (patch bond, downstream crossing all attention layers) and on an exact feed-forward classifier:

substrate	ranking	deletion AUC ↓	insertion AUC ↑	
χ -ViT (attn)	ODT (ours)	0.150	0.792	(3 / 2 seeds; std $\leq$ 0.03)
	PCA	0.151	0.788	
	random	0.498	0.491	
classifier	ODT (data-free)	0.222	0.786	
	integrated gradients	0.191	0.784	
	PCA	0.204	0.800	
	random	0.548	0.530	

The ODT ranking is strongly faithful — it crushes random — and is *competitive with*, not superior to, PCA and integrated gradients. Its distinctive properties are that it is *weight-derived* (at feed-forward bonds it needs no data at all, unlike PCA and integrated gradients) and *globally rankable* (integrated gradients is per-sample). Through attention we fall back to a data-driven output-sensitivity Gram, so the “for free” claim is honest only at feed-forward bonds; the causal structure nonetheless survives attention (the χ -ViT bond above crosses four attention layers) and, in our small models, appears to grow *more* compressible with depth (Section 4.5).

5.2 Coherence is bought by topology, at no capability cost

Whether the causal directions are *human-coherent* (localized, nameable) is not a property of convertibility — it is set by architecture topology. On single-bilinear classifiers whose class logit is exactly a quadratic form, we measure eigen-atom spatial locality (energy in the top 10% of pixels, relative to random directions), across three seeds and two datasets:

architecture	SVHN		CIFAR-10		
	acc	locality ($\times$ rand)	acc	locality ($\times$ rand)	
dense	0.747	2.29	0.354	1.47	(mean over 3 seeds, std $\leq$ 0.05)
conv (spatial readout)	0.814	3.45	0.630	3.42	
locally-connected	0.800	3.48	0.584	3.44	

Spatial-locality-preserving topology reaches $\sim 3.5\times$ random-baseline locality on both datasets — versus 2.29 (SVHN) / 1.47 (CIFAR) for dense, i.e. $\sim 1.5\text{--}2.3\times$ more localized than dense — and the gap is *larger* on the harder dataset. The effect is not a capacity confound. A width sweep ($8 \rightarrow 128$) shows dense locality is *capacity-independent*: growing the dense model raises accuracy ($0.70 \rightarrow 0.75$) but its atom locality stays flat at ~ 2.2 , never approaching the ~ 3.5 of the structured topologies at any width — so coherence is set by topology, not parameters or accuracy (and the most coherent model is also the most accurate and parameter-efficient). A deep-network replication (data-driven input-space ranking, depth 1 vs 3) shows the gap survives depth and *strengthens* with it (conv locality $2.05 \rightarrow 2.52$ on CIFAR while dense stays ~ 1.4). One honest subtlety: plain convolution with *global* pooling underperforms — global pooling imposes translation-invariance that both starves capacity and delocalizes modes; the spatial readout recovers both.

5.3 For a policy, mechanisms live behind the attention

On χ -VLA (the synthetic reach task), the action-relevant quantity is which object the instruction names and where it is. This must be *constructed* by attention (bind the named identity to a location), and indeed it is not linearly present at the network input but is at the post-attention action-query bond:

quantity (linear decodability)	input bond	post-attention bond
target position x/y (R^2 , 3 seeds)	0.20 $\pm$.01	0.92$\pm$.03
gripper position (in the state token)	0.99	0.97
named colour / shape (in the instruction)	1.00	0.94

Only the *bound* quantity (target position) jumps across the attention; the raw ingredients are linear at both bonds. We then verify the mechanism is *causal* with an intervention: rewriting the instruction to name a different present object (image and state fixed) makes the predicted action follow the new target, with $R^2 = 0.956 \pm 0.016$ and a 97.1% switch-rate (3 seeds). Notably a purely *correlational* handle fails — patching the linear target-position *decode* direction does not steer the action (transfer 0.002 vs 0.012 for random), an instance of the decode-is-not-control gap [18] that motivates causal over correlational interpretation.

6 The substrate is a genuine, competitive tensor network

Under a hard compute limit (consumer GPUs; small models, modest data — a verified demonstration, not a scale record), we confirm the interpretability findings are computed on a real, competitive substrate, not a degenerate one. The foldable χ -ViT reaches 98.8% of a parameter-matched softmax baseline on SVHN (0.917 vs 0.928); a cross-bilinear projector is worth +81 accuracy points over concatenation on a task whose label is, by construction, a product of two streams, and zeroing the product term erases exactly that gain (evidence the explicit multiplication does real work); and an end-to-end χ -VLA policy uses language measurably — $22\times$ better than a language-blind baseline on a synthetic reach task, and a $\sim 9\times$ shuffled-instruction grounding gap on real LIBERO-Object behaviour cloning (offline action error only; closed-loop control is out of scope). The 450M reference configuration is specified and its size verified (448.2M) but untrained, for lack of data-center hardware.

7 Related work

Weight-based / tensor interpretability. Our substrate is the bilinear/tensor line: bilinear MLPs enable weight-based mechanistic interpretability by reading a class output as a quadratic form whose eigenvectors are features [5, 2]; Doot et al. [3] give tree-structured orthogonalize–diagonalize–truncate (ODT) as a *compression* result; Déhérand [4] extends the eigen-feature analysis to convolutional models and observes that convolutional eigenvectors look more coherent. We depart from all three by separating *causal* from *coherent* as an explicit, intervention-tested axis, by isolating topology as the coherence lever at matched capability with depth-scaling, and by the bond-placement principle for policies.

Mechanistic interpretability, features, and dictionaries. A large body studies features and circuits in trained networks [6, 7] and the superposition of features in polysemantic directions [8]. The dominant current tool for extracting a feature basis is sparse dictionary learning over activations (sparse autoencoders) [9, 10, 11]. Those methods learn an *over-complete, data-driven* dictionary; our ODT basis is instead *exact and weight-derived* (at feed-forward bonds, data-free) and *globally rankable* — a complementary object, which we show is as faithful as an unsupervised variance basis.

Attribution, causal mediation, and probing. We benchmark against integrated gradients [12] and compare our global ranking to activation-variance (PCA) bases, using the deletion/insertion faithfulness protocol [13, 14]. Our causal claims connect to causal mediation /

activation patching [15, 16]; our steering *null* (a linear decode direction is not a control direction) is an instance of the known gap between probing and causation [17, 18].

Tensor methods and polynomial networks. Bilinear layers are CP/Tucker factorizations [19]; tensor networks have a long history in ML [20], and polynomial (II-)networks show such models are competitive [21]. **Vision-language-action models.** Our policy substrate follows the modular VLA line [1, 22, 23]; VLA interpretability is nascent, and a causal, weight-derived action-mechanism basis is, to our knowledge, new.

8 Discussion and limitations

By construction the χ -VLA forward map is a tensor network, and we verify this end to end for the language backbone, a one-layer and a two-layer attention model, and the feed-forward decomposition model. The central new finding is that global low-rank structure, the property that makes ODT worthwhile, is present not only in feed-forward chains but also through attention and residual connections, and it grows rather than shrinks with depth in our small models.

The honest boundaries are as follows. The exact, weight-only, data-free version of global ODT is established here only for feed-forward downstream bonds; for attention-crossing bonds we currently use a data-driven importance measure, and the transfer of Dooms et al.’s error bound to attention networks is not proven. The strict foldable model is stable and matches its non-strict teacher only in the small-residual-gain regime, which is a real constraint we did not need to pay any accuracy for at this scale but which may matter at larger scale. All experiments are small; the reference-scale model is specified and its size verified but not trained, for lack of data-center hardware. On the robot task we report offline action error and a language-grounding gap, not closed-loop success.

The interpretation results also point past the current architecture. Our atoms are causal but only mildly localized, and Déhérand’s convolutional comparison [4] shows why: the tensor-network *topology* decides whether the spectral modes are human-coherent, not the mere fact of tensor-convertibility. This suggests the next χ -VLA should build coherence into its structure rather than hope to read it out afterwards — for instance local or multiscale bilinear vision cores in place of dense ones, fixed structural tensors that separate world and robot geometry from learned policy, symmetry-aware canonicalization that groups translated copies of a mechanism into one orbit, and an output-conditioned, modality- and degree-resolved decomposition of continuous actions (which signed mechanism drives “move left”, and does it survive intervention). Déhérand also warns that a convertible network need not be tractably *contractible*; its unrolled tensor network can grow exponentially through fan-out, which argues for bounded-treewidth design (local attention, object-slot bottlenecks) from the start.

9 Conclusion

The parts needed for a fully tensor-decomposable VLA already exist, and OpenVLA’s modular design is ready to receive them. Assembling bilinear feed-forward layers, softmax-free bilinear attention, foldable normalization, and a linear action head gives χ -VLA, a policy that is a single tensor network from pixels and token identities to continuous actions. We showed it folds exactly, stays competitive after folding, benefits measurably from the explicit cross-stream product, uses language on a real robot dataset, and, uniquely among policy architectures, admits an exact global diagonalization at feed-forward bonds in which most hidden dimensions can be removed and a global importance ranking beats a local one; through attention the same ranking (computed data-drivenly) continues to beat local and random baselines, though the exact weight-only contraction

across attention remains open.

A Making the foldable model stable

Whether the model is a tensor network is settled by construction; the subtle question is whether the foldable (scalar) normalization is stable inside a deep stack. Per-token normalization keeps activations at unit scale on every input but its divisor is input-dependent, so it cannot be folded; a frozen scalar c is foldable but cannot rescale an input that is larger or smaller than average. Because the bilinear layer is degree two, an input at twice the average scale produces an update about four times too large, and the error grows with depth. The resolution is to keep residual branches near the identity: a small learned gain per branch (initialized 0.01) so the degree-two amplification never compounds, trained by distilling a stable per-token teacher into a scalar-normalization student. On a language model the student matches its teacher (val loss 5.04 vs 5.0) and folds to a normalization-free graph with error $\sim 10^{-6}$ (fp32) / 10^{-12} (fp64). Empirically the largest per-token magnitude at any normalization site is only $\sim 2.8\times$ the average across 1.6M tokens and does not grow with depth, so the fixed scalar is an accurate stand-in.

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Figure 1: OpenVLA (left) versus χ -VLA (right), stage by stage. Green boxes are the polynomial (tensorized) replacements; blue boxes are the non-polynomial originals they replace; grey boxes are kept unchanged or lie outside the tensor-network boundary. Patch embedding and the tokenizer are already linear and need no change. The softmax, smooth-activation, and layer-normalization blocks (the dual vision encoders, the MLP projector, and the Llama 2 backbone) and the discrete action decoder are each swapped for a polynomial equivalent with foldable normalization. Camera decoding, string tokenization, and safety clipping stay external.

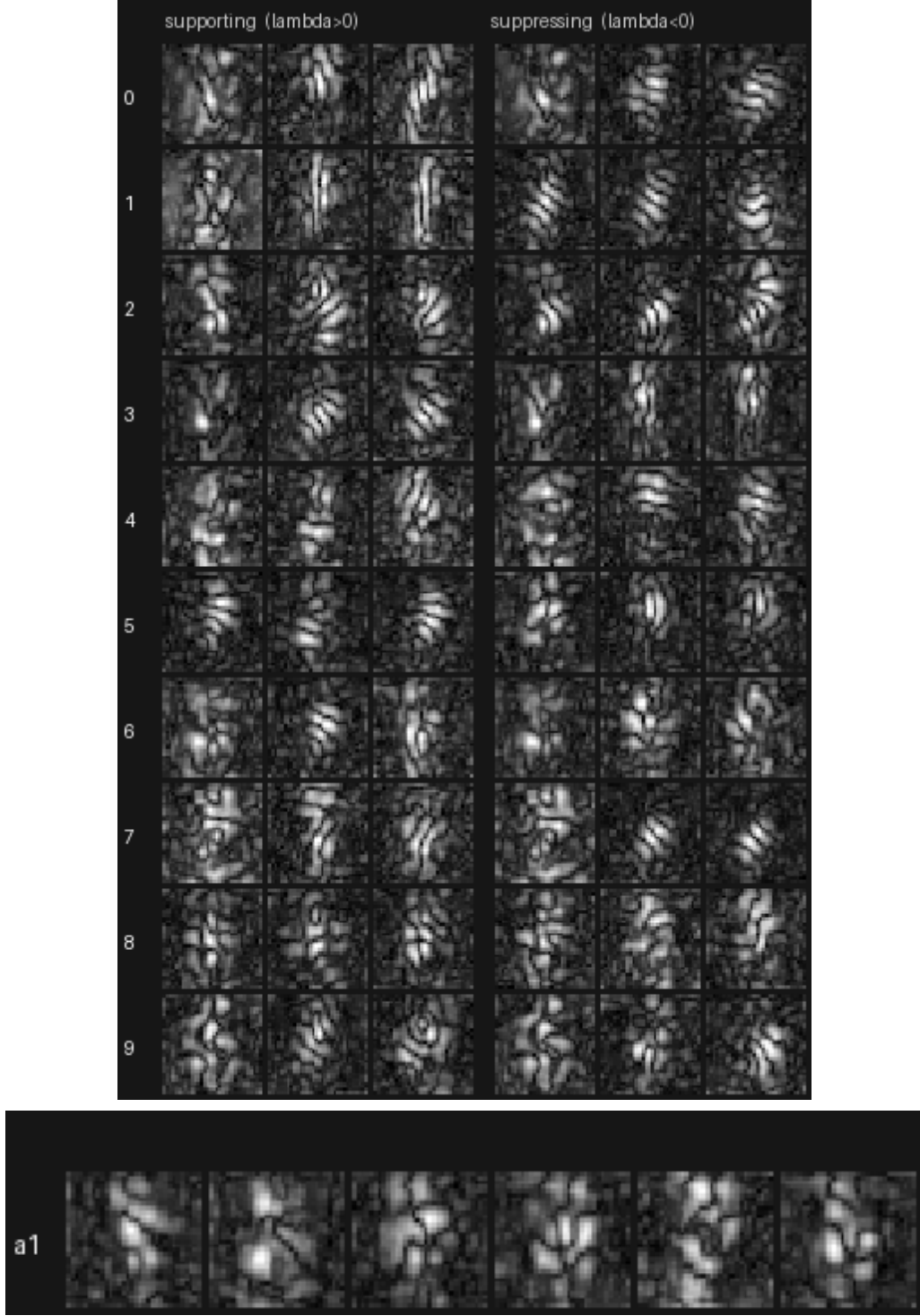


Figure 2: Extracted ODT eigen-atoms projected to pixels (brighter = higher sensitivity). **Top:** the single-bilinear χ -classifier, one row per SVHN class; the three left atoms of each row *support* the class ($\lambda > 0$), the three right *suppress* it ($\lambda < 0$). **Bottom:** the six top global directions of the deep three-layer χ -MLP. The atoms are causal and class-relevant; how *localized* they are depends on architecture topology, which we quantify across dense/convolutional/locally-connected models in Section 5.